

Measurement-Driven Latency Analysis of Uplink/Downlink Scheduling in DECT-2020 NR

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Abstract—Digital Enhanced Cordless Telecommunications - 2020 New Radio (DECT-2020 NR) is an European Telecommunications Standards Institute (ETSI)-standardized non-cellular radio technology targeting low-latency communication in private and enterprise-grade wireless networks. While prior experimental studies have reported promising end-to-end latency figures, the contribution of Uplink (UL)/Downlink (DL) scheduling behavior and hardware-induced Receive (RX)-to-Transmit (TX) (RX-TX) turnaround to the overall latency budget has not been thoroughly quantified in practical implementations.

In this work, we present a measurement-driven latency analysis of UL/DL scheduling behavior in DECT-2020 NR star networks. Using Digital Storage Oscilloscope (DSO) measurements on a single-chip DECT-2020 NR platform, complemented by internal modem timing analysis, we characterize the RX-TX turnaround delay associated with UL/DL alternation and evaluate its impact on end-to-end latency under different traffic and scheduling scenarios. The results show that, for short-packet and request-response traffic, turnaround-induced latency can constitute a significant portion of the total delay and is strongly influenced by scheduling decisions. Based on these observations, we discuss the limitations of scheduling-based mitigation on single-transceiver hardware and outline architectural directions, such as UL/DL channel separation, to further reduce latency in future DECT-2020 NR systems.

Index Terms—DECT-2020 NR, latency analysis, UL/DL scheduling, Time Division Duplexing (TDD) systems, RX-TX turnaround, star topology, measurement-based evaluation

I. INTRODUCTION

Digital Enhanced Cordless Telecommunications -2020 New Radio (DECT-2020 NR) is a non-cellular 5G radio technology standardized by ETSI to support Ultra-Reliable Low-Latency Communication (URLLC) and massive Machine-Type Communication (mMTC) use cases in private and industrial wireless networks [?], [?], [?]. It operates in harmonized, license-exempt spectrum allocated for DECT-2020 NR-compliant devices, enabling predictable interference conditions without requiring individual operator licenses [?], [?].

Recent experimental studies have demonstrated that DECT-2020 NR can achieve sub-millisecond to few-millisecond end-to-end latency under favorable conditions [?], [?], [?]. These results confirm the suitability of DECT-2020 NR for latency-sensitive applications such as industrial automation, tactile Internet services, and real-time audio transmission. However, most existing evaluations report latency as an aggregate end-

to-end metric and provide limited insight into the contribution of underlying radio operation and scheduling behavior.

In practical DECT-2020 NR implementations, cost and power constraints often motivate the use of a single transceiver chain in which Uplink (UL) and Downlink (DL) communication share the same Radio Frequency (RF) and baseband hardware [?]. Since DECT-2020 NR relies on Time Division Duplexing (TDD), UL and DL transmissions cannot occur simultaneously, and each change in transmission direction requires a finite transceiver reconfiguration interval. Although the standard defines slot-based guard periods to support UL/DL transitions [?], the resulting Receive (RX)-Transmit (TX) turnaround time is implementation-dependent and may introduce non-negligible latency, particularly for short-packet and request-response traffic patterns.

Although scheduling and UL/DL coordination have been extensively studied in TDD-based cellular systems to support URLLC traffic [?], comparatively little attention has been given to implementation-level turnaround effects in DECT-2020 NR systems. Existing DECT-2020 NR studies primarily rely on simulations or coarse-grained measurements and rarely decompose the contribution of RX-TX switching delay or its interaction with scheduling behavior [?], [?]. As a result, the extent to which observed latency is fundamentally limited by hardware constraints rather than scheduling decisions remains insufficiently characterized.

This paper presents a measurement-driven analysis of UL/DL latency in DECT-2020 NR star networks. Using Digital Storage Oscilloscope (DSO)-based measurements on a single-chip DECT-2020 NR platform, complemented by internal modem timing analysis, we characterize the RX-TX turnaround time and evaluate its contribution to end-to-end latency. We further analyze how scheduling behavior influences the frequency and accumulation of direction switching and discuss the extent to which scheduling-level optimizations can mitigate turnaround-induced latency. The goal of this work is not to propose a new scheduling algorithm, but to provide empirical insight into latency bottlenecks arising from implementation-specific hardware and firmware constraints that are not mandated by the DECT-2020 NR standard.

II. BACKGROUND AND RELATED WORK

A. DECT-2020 NR Standardization and Design Goals

The European Telecommunications Standards Institute (ETSI) standards define a TDD physical layer with a slot-based frame structure and guard periods to support transitions between UL and DL operation [?], [?]. While these mechanisms are specified at the protocol level, implementation aspects such as transceiver reconfiguration latency and scheduling behavior are intentionally left open to vendors.

B. Experimental Evaluation of DECT-2020 NR Latency

Recent experimental studies have demonstrated that DECT-2020 NR is capable of achieving low end-to-end latency under favorable conditions. Practical evaluations in the context of professional live audio transmission report sub-millisecond to few-millisecond latency using DECT-2020 NR hardware [?]. Similarly, work targeting tactile Internet and human-in-the-loop applications highlights the suitability of DECT-2020 NR for latency-critical communication and emphasizes its potential for URLLC-like operation [?].

Additional experimental and link-level evaluations further report low round-trip times and reliable communication performance [?], [?]. These studies provide valuable insight into achievable latency figures and overall system behavior. However, latency is typically reported as an aggregate metric, without detailed decomposition into scheduling delay, transceiver turnaround time, or other implementation-level contributors.

C. Scheduling and UL/DL Coordination

UL and DL scheduling have been widely studied in TDD-based wireless systems, particularly in the context of 5G cellular networks supporting URLLC traffic. Prior work shows that frame coordination and scheduling decisions play a crucial role in minimizing latency and avoiding excessive direction switching overhead [?]. These studies demonstrate that even when physical-layer transmission times are short, scheduling-induced delays can dominate end-to-end latency.

In DECT-2020 NR systems, scheduling and resource allocation are intentionally flexible to support a wide range of deployment scenarios [?]. System-level DECT-2020 NR studies primarily focus on routing, scalability, and multi-hop performance in mesh topologies [?], [?]. While these works address important networking aspects, they do not explicitly analyze the impact of UL/DL alternation and transceiver turnaround on latency in star-based or single-hop deployments.

III. SYSTEM MODEL AND EXPERIMENTAL SETUP

This section describes the considered system model, network topology, and hardware assumptions used throughout this work. The focus is on a practical industrial deployment of DECT-2020 NR, where a central controller coordinates latency-sensitive sensor and actuator traffic.

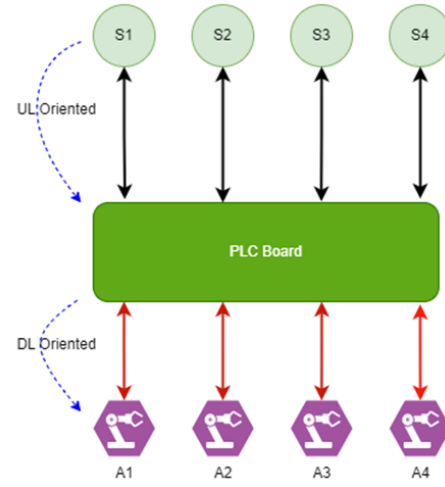


Fig. 1. Industrial sensor-actuator communication model using a DECT-2020 NR star topology. Sensors (S1–S4) transmit data to a central PLC in the UL, while actuators (A1–A4) receive commands in the DL.

A. Industrial Sensor-Actuator Star Topology

We consider an industrial sensor-actuator network organized in a star topology, as illustrated in Fig. ???. The network consists of a programmable logic controller (PLC) acting as the DECT-2020 NR Fixed Termination node (FT node), and multiple Portable Termination node (PT node) attached to sensors (S1–S4) and actuators (A1–A4). All wireless communication is coordinated by the FT node, and direct communication between PT nodes is not supported.

Sensor PT nodes periodically or event-wise transmit measurement data to the PLC in the UL direction, while actuator PT nodes receive control commands from the PLC in the DL direction.

B. Duplexing and Communication Model

DECT-2020 NR operates using TDD, in which UL and DL transmissions share the same frequency resources and are separated in time. As a result, the system alternates between UL-oriented phases, dominated by sensor data transmissions, and DL-oriented phases, dominated by actuator command delivery.

In the considered system model, the PLC /FT node cannot transmit and receive simultaneously. Instead, it switches between RX and TX states depending on the current scheduling decision. Guard intervals are used to support these transitions; however, the duration of the RX–TX switching process is implementation-dependent and not explicitly defined by the DECT-2020 NR standard.

C. Scheduling Scope and Latency Components

This work does not assume a specific scheduling algorithm; instead, it analyzes the observable scheduling behavior and its interaction with RX–TX switching under different UL/DL transmission scenarios. Higher-layer protocols, multi-hop routing, and mesh networking are outside the scope of this study.

Based on the above assumptions, the end-to-end latency experienced by sensor and actuator traffic can be decomposed into three main components: (i) scheduling and queuing delay at the Medium Access Control (MAC) layer, (ii) RX–TX turnaround delay caused by transceiver reconfiguration, and (iii) physical-layer transmission time. The primary focus of this paper is on quantifying the turnaround-related component and evaluating how UL/DL scheduling decisions influence its impact on latency.

D. Experimental Setup and Hardware Platform

Fig. ?? shows the complete laboratory measurement setup used in this study. Experiments were conducted using two DECT-2020 NR nodes based on the Nordic Semiconductor nRF9151 system-on-chip. One node was configured as the DECT-2020 NR (Fixed Termination (FT)), acting as the central controller, while the other node was configured as a (Portable Termination (PT)). The nRF9151 integrates the radio-frequency front-end, baseband processing, and protocol stack on a single device.

In this platform, transmission and reception share a single RF transceiver chain, preventing simultaneous RX and TX operation. Consequently, each transition between UL and DL communication requires a hardware and firmware re-configuration of the transceiver, resulting in a finite RX–TX turnaround delay. This delay is influenced by the internal radio state machine, firmware processing, and timing alignment with the DECT-2020 NR frame structure.

To measure this turnaround delay, GPIO timing markers generated by the firmware were captured using a DSO. During the measurements, the LTE/DECT-2020 NR RF ports were connected to laboratory instruments to monitor RF activity and, in some runs, to provide a controlled termination. As a result, radiated power was strongly attenuated, and the two nodes communicated only at very short range, primarily via leakage coupling around the RF connectors and nearby cables. The setup was designed to study timing and scheduling effects rather than coverage or link-budget performance.

The nodes were operated using the `dect_shell` DECT-2020 NR reference application with minimal instrumentation for timing extraction.

E. Instrumentation and Timing Markers

To obtain ground-truth timing measurements, GPIO signals were generated by the firmware to indicate key radio events. Specifically, GPIO markers were asserted upon completion of reception and upon initiation of transmission at the FT. These signals were connected to a DSO, enabling direct measurement of the RX–TX turnaround delay.

In addition to external measurements, internal modem timing information was collected using firmware-accessible timers provided by the DECT-2020 NR modem. These timers report operation latencies in internal modem ticks and provide insight into radio state transitions and internal processing behavior.

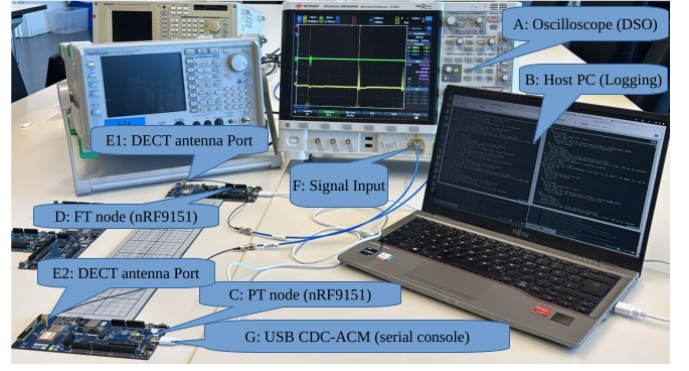


Fig. 2. Laboratory setup for measuring RX–TX turnaround in DECT-2020 NR using two nRF9151 nodes and GPIO timing markers captured by a DSO.

F. Measurement Procedure

Measurements were performed under controlled conditions with minimal background traffic. The FT and PT exchanged UL and DL traffic according to predefined scheduling patterns. For each experiment, multiple RX–TX transitions were captured to verify repeatability and consistency. The measured turnaround delay was extracted as the time difference between the RX completion marker and the TX start marker observed on the DSO.

IV. MEASUREMENT RESULTS AND SCHEDULING IMPACT

This section analyzes how UL scheduling behavior interacts with the measured RX–TX turnaround delay and how this interaction affects end-to-end latency.

A. Scheduling Scenarios

To analyze the impact of scheduling on latency, two representative UL-oriented scheduling scenarios are considered. In Scenario 1, UL transmissions are scheduled sequentially across frames, with each UL transmission followed by an acknowledgment in the DL direction. This results in frequent UL/DL alternation and repeated RX–TX turnaround events.

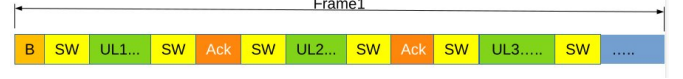
In Scenario 2, multiple UL transmissions are grouped within a single frame to amortize scheduling overhead. While this reduces frame-level coordination overhead, acknowledgment handling still requires repeated RX–TX switching, such that turnaround-induced latency remains dominant.

Fig. ?? shows the measured end-to-end latency as a function of the number of UL transmissions for both scenarios. In both cases, latency increases approximately linearly, which is expected since each additional UL transmission incurs an approximately constant RX–TX turnaround delay on the single-transceiver platform.

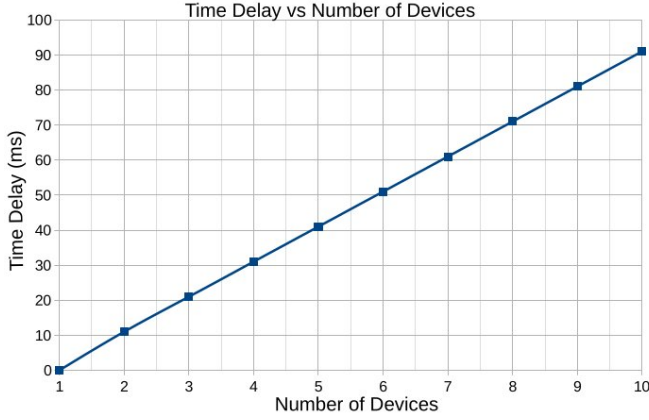
This behavior can be explained analytically using the slot-based frame structure of DECT-2020 NR. For Scenario 1, each UL transmission incurs a fixed overhead of approximately seven slots (2.9 ms), in addition to the UL data transmission time. The resulting normalized latency $L_n^{(1)}$,



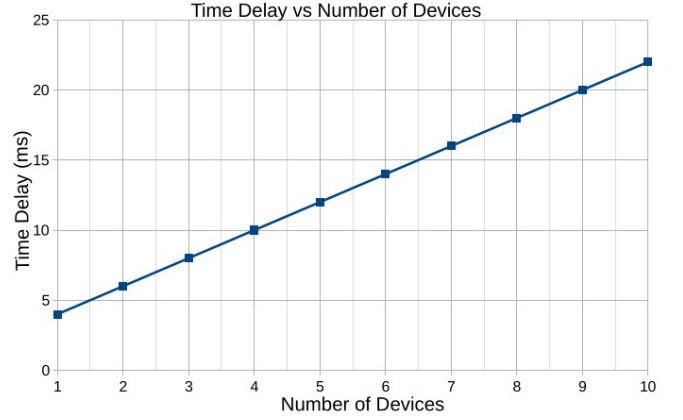
(a) Scenario 1: Sequential UL scheduling



(b) Scenario 2: Grouped UL scheduling



(c) Measured latency for Scenario 1



(d) Measured latency for Scenario 2

Fig. 3. UL scheduling scenarios and their impact on turnaround-induced latency. (a) Sequential UL transmissions with acknowledgments after each packet. (b) Grouped UL transmissions with acknowledgments after a batch. (c) Measured latency versus number of UL transmissions for Scenario 1. (d) Measured latency versus number of UL transmissions for Scenario 2.

normalized with respect to the 10 ms frame duration, can be expressed as

$$L_n^{(1)} = \frac{7 \text{ slots} + T_{\text{data}}^{\text{UL}_x}}{10 \text{ ms}} = \frac{2.9 \text{ ms} + T_{\text{data}}^{\text{UL}_x}}{10 \text{ ms}}, \quad (1)$$

where $T_{\text{data}}^{\text{UL}_x}$ denotes the payload transmission time of a single UL packet. As shown in Eq. (1), the latency consists of a constant scheduling overhead and a variable payload transmission component.

In Scenario 2, the scheduling overhead is shared across multiple UL transmissions. In the considered configuration, a batch of three UL transmissions incurs a fixed overhead of approximately nineteen slots (7.9 ms), while the combined UL payload occupies four slots (1.67 ms). The corresponding normalized latency $L_n^{(2)}$ is given by

$$L_n^{(2)} = \frac{19 \text{ slots} + T_{\text{data}}^{\text{UL}_{1,2,3}}}{10 \text{ ms}} = \frac{7.9 \text{ ms} + T_{\text{data}}^{\text{UL}_{1,2,3}}}{10 \text{ ms}}, \quad (2)$$

with $T_{\text{data}}^{\text{UL}_{1,2,3}} = 4 \text{ slots} \approx 1.67 \text{ ms}$. Equation (2) illustrates that grouping multiple UL transmissions allows the scheduling overhead to be amortized across several packets.

These results show that scheduling alone provides limited mitigation of turnaround-induced latency in practical DECT-2020 NR implementations, as the dominant contribution arises from unavoidable RX–TX switching.

B. Turnaround-Induced Latency and Baseline Scheduling Behavior

Based on the measurements, the RX–TX turnaround latency can be conceptually decomposed into three main components: (i) internal RX processing and completion,

(ii) transceiver reconfiguration and TX initiation, and (iii) firmware execution and alignment with the DECT-2020 NR slot structure. While the exact contribution of each component is implementation-dependent, the measurements indicate that no single component can be neglected when analyzing low-latency communication behavior.

In the considered DECT-2020 NR implementation, UL and DL transmissions are scheduled sequentially using TDD. Each change in transmission direction requires either a RX–TX or TX–RX transition at the FT or PT, thereby incurring the measured turnaround delay. As a result, the scheduling behavior directly determines how frequently this composite turnaround latency is accumulated over time and, consequently, how strongly it impacts end-to-end latency.

C. RX–TX Turnaround Measurement

Fig. ?? illustrates the measured RX–TX turnaround process using GPIO timing markers captured on a DSO. The yellow signal indicates the UL transmission initiated by the PT, during which the FT is actively receiving. Once the UL reception is completed, the FT must reconfigure its transceiver from reception to transmission mode before it can respond. This reconfiguration results in a clearly observable idle interval between the end of the UL transmission and the start of the DL response. The measured RX–TX turnaround delay is approximately 850 μs , corresponding to nearly two DECT-2020 NR slots. Although the over-the-air transmission time is included in the measurement, it is negligible due to the very short separation (approximately 30 cm) between the nodes, such that the observed delay is dominated by transceiver

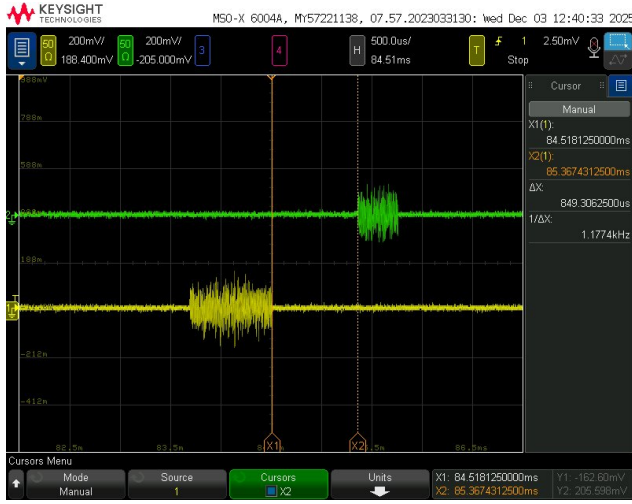


Fig. 4. RX–TX turnaround behavior at the FT measured using GPIO timing markers. The yellow trace shows the UL transmission from the PT, while the green trace indicates the DL response after the turnaround interval.

reconfiguration, firmware processing, and frame alignment effects.

Repeated measurements across multiple RX–TX transitions show consistent turnaround values with limited variation, indicating stable behavior under the considered operating conditions. The magnitude of the measured delay highlights that RX–TX switching constitutes a non-negligible component of the end-to-end latency in practical DECT-2020 NR systems.

D. Cross-Validation Using Internal Modem Timing

To further interpret the DSO-based measurements, internal modem timing information was analyzed. The DECT-2020 NR modem reports operation latencies in internal modem ticks, where one DECT-2020 NR slot corresponds to 28 800 ticks. These timers capture the duration of internal radio state transitions and baseband processing steps, providing insight into delays that are not directly observable at the RF interface.

The reported modem timings indicate that reception-related processing, including RX completion and RSSI handling, requires several hundred microseconds, while the TX initiation procedure alone spans approximately one DECT-2020 NR slot. When combined with firmware execution and slot-alignment effects, these internal delays are consistent with the approximately 850 μ s RX–TX turnaround measured using the DSO. This agreement confirms that the observed turnaround latency is dominated by transceiver reconfiguration and scheduling-related effects rather than measurement artifacts.

Since the RX–TX turnaround process is governed by deterministic transceiver state transitions and frame alignment, repeated measurements under identical conditions exhibit only negligible variation. Consequently, the analysis focuses on systematic turnaround behavior rather than statistical dispersion.

V. DISCUSSION, DESIGN IMPLICATIONS, AND FUTURE DIRECTION

This section discusses the implications of the measurement results for UL/DL scheduling and hardware design in practical DECT-2020 NR systems. The focus is on understanding the fundamental limitations observed in the experiments and identifying directions for latency reduction beyond scheduling-level optimizations.

A. Limits of Scheduling-Based Latency Reduction

The results presented in the previous sections show that RX–TX turnaround delay constitutes a dominant component of end-to-end latency in the considered DECT-2020 NR implementation. Both DSO measurements and internal modem timing indicate that each change in transmission direction incurs a delay on the order of multiple DECT-2020 NR slots. As a result, scheduling decisions that frequently alternate between UL and DL inherently accumulate significant latency.

The analysis of representative scheduling scenarios demonstrates that while grouping UL transmissions can reduce some scheduling overhead, it does not eliminate the need for RX–TX switching when acknowledgments or DL responses are required. Consequently, scheduling strategies alone offer limited potential for reducing latency in traffic patterns characterized by short packets or request–response behavior, which are common in industrial sensor–actuator applications.

B. Impact on Industrial Sensor–Actuator Traffic

In the considered sensor–actuator star topology, UL sensor data and DL actuator commands are tightly coupled in time. This asymmetry naturally leads to frequent UL/DL alternation at the FT. The measured turnaround delay, therefore, directly affects control-loop latency and limits how quickly sensor information can be translated into actuator actions.

For such use cases, the observed latency is not primarily determined by physical-layer transmission time, but by the overhead associated with transceiver reconfiguration and scheduling alignment. This finding highlights the importance of considering hardware-induced delays when evaluating the suitability of DECT-2020 NR for low-latency industrial control applications.

C. Architectural Considerations and Future Directions

The measurement results presented in this work indicate that RX–TX turnaround delay is a dominant contributor to end-to-end latency in practical DECT-2020 NR systems employing single-transceiver hardware. This section discusses architectural considerations that arise from these findings and outlines potential directions for reducing turnaround-induced latency in future DECT-2020 NR implementations.

1) Single-Transceiver Versus Multi-Chain Architectures:

The evaluated nRF9151-based platform employs a single RF transceiver chain shared between reception and transmission. While this architecture offers advantages in terms of cost, power consumption, and implementation complexity, it inherently requires sequential RX and TX operation. As shown

by the DSO measurements and modem timing analysis, the resulting RX–TX turnaround delay spans multiple DECT-2020 NR slots and cannot be eliminated through scheduling alone.

One architectural direction for reducing this limitation is the introduction of separate receive and transmit chains. Even partial separation, such as independent baseband processing with shared RF components, could reduce reconfiguration overhead and enable faster direction switching. However, such approaches introduce trade-offs related to hardware complexity, power consumption, and device cost, which must be carefully evaluated for industrial deployments.

2) *UL and DL Resource Separation*: Another potential approach to mitigating turnaround-induced latency is the separation of UL and DL resources in frequency or time. Allocating distinct channels for UL and DL communication could reduce or eliminate the need for frequent RX–TX switching at the FT, particularly in sensor–actuator scenarios with tightly coupled request–response traffic.

While DECT-2020 NR primarily relies on TDD to efficiently utilize the harmonized, license-exempt spectrum allocated for DECT-2020 NR systems, future system designs could explore hybrid approaches that trade spectral efficiency for reduced latency in selected operating modes. Such mechanisms could be activated dynamically for latency-critical traffic, while preserving conventional TDD operation for best-effort communication.

3) *Scheduling Awareness of Turnaround Costs*: The results of this study also suggest that scheduling decisions should explicitly account for hardware-induced turnaround costs. Rather than assuming negligible switching overhead, schedulers could incorporate turnaround-aware metrics when selecting transmission directions and grouping traffic. While such awareness cannot remove the turnaround delay itself on single-transceiver platforms, it may help avoid unnecessary direction changes and reduce latency accumulation under certain traffic patterns.

Importantly, the effectiveness of turnaround-aware scheduling is fundamentally bounded by the underlying hardware architecture. As demonstrated in this work, scheduling optimization alone offers limited gains when turnaround delays dominate the latency budget.

4) *Measurement-Driven System Design*: Finally, the combination of external DSO measurements and internal modem timing analysis highlights the value of measurement-driven system design. External measurements provide ground-truth latency values observable at the RF interface, while internal timing information helps explain the origin of these delays. Future DECT-2020 NR system evaluations would benefit from similar multi-level measurement approaches to accurately characterize the interaction between hardware, firmware, and scheduling behavior.

VI. CONCLUSION

This paper presented a measurement-driven analysis of UL/DL scheduling and its impact on latency in DECT-2020

NR systems, based on DSO-based timing measurements performed on an nRF9151 platform. The RX–TX turnaround delay was quantified at approximately two DECT-2020 NR slots and identified as a dominant contributor to end-to-end latency, particularly for short-packet and request–response traffic. Scenario-based evaluations showed that although scheduling decisions influence how frequently turnaround delays occur, they offer limited potential for latency reduction on single-transceiver architectures. In industrial sensor–actuator use cases with frequent UL/DL alternation, latency accumulates in a manner that cannot be effectively mitigated through scheduling alone, highlighting the need to account for hardware-induced delays in low-latency DECT-2020 NR system design.

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